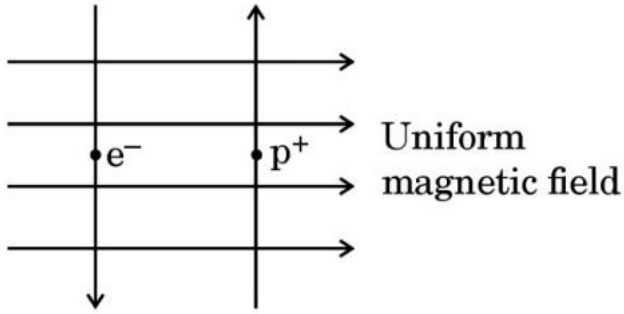
SECTION C

30. A uniform magnetic field exists in the plane of paper as shown in the diagram. In this field, an electron (e^-) and a positron (p^+) enter as shown. The electron and positron experience forces :



(1)

(A) both pointing into the plane of the paper.
(B) both pointing out of the plane of the paper.
(C) pointing into the plane of the paper and out of the plane of the paper respectively.
(D) pointing out of the plane of the paper and into the plane of the paper respectively.

Ans: A

31. The colour of light for which the refractive index is minimum is (1)

(A) Red
(B) Yellow
(C) Green
(D) Violet

Ans: A

32. Assertion (A): Electrons move from lower potential to higher potential in a conductor.
Reason (R): A dry cell maintains electric potential difference across the ends of a conductor. (1)

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).
(B) Both Assertion (A) and Reason (R) are true, but Reason (R) is not the correct explanation of the Assertion (A).
(C) Assertion (A) is true, but Reason (R) is false.
(D) Assertion (A) is false, but Reason (R) is true.

Ans: B

33. An electric source can supply a maximum charge of 500 coulomb. If the current drawn by a device is 25 mA, find the time in which the electric source will be discharged completely. (2)

Ans: Current, charge and time are related by the relation; $I=\frac{Q}{t}$, **or** $t=\frac{q}{I}=\frac{500}{0.025}=2\times10^4 s$

34. (A) A student fixes a white sheet of paper on a drawing board. He places a bar magnet in the centre and sprinkles some iron filings uniformly around the bar magnet. Then he taps gently and observes that iron filings arrange themselves in a certain pattern.

(a) Why do iron filings arrange themselves in a particular pattern?

(b) Which physical quantity is indicated by the pattern of field lines around the bar magnet? (2)

OR

(B) A compass needle is placed near a current carrying wire. State your observations for the following cases and give reasons for the same in each case-

(a) Magnitude of electric current in wire is increased.

(b) The compass needle is displaced away from the wire. (2)

Ans: (A) (a) When iron filings are placed in a magnetic field around a bar magnet, they behave like tiny magnets. The magnetic force experienced by these tiny magnets make them rotate and align themselves along the direction of field lines.

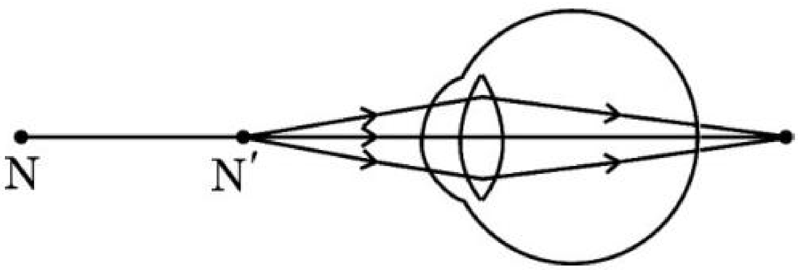
(b)The physical property indicated by this arrangement is the magnetic field produced by the bar magnet.

OR

(B) (a) The deflection in the compass needle increases as Magnetic field of the current carrying conductor is directly proportional to current flowing through it.

(b) The deflection in the needle decreases as the magnetic field is inversely proportional to the perpendicular distance from the wire.

35. (A) Study the diagram given below and answer the questions that follow :



(i) Name the defect of vision depicted in this diagram stating the part of the eye responsible for this condition.

(ii) List two causes of this defect.

(iii) Name the type of lens used to correct this defect and state its role in this case. (3)

OR

(B) What do you mean by dispersion of white light ? State its cause. Draw a diagram to show dispersion of a beam of white light by a glass prism. (3)

Ans: (A) (i) Hypermetrompia or long sightedness, eye lens

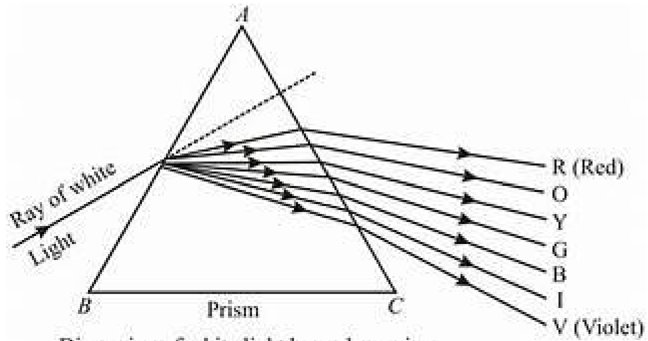
(ii) Lesser rate of growth of eye balls, loss of flexibility of eyelens (any other logical cause)

(iii) convex lens, converges the beam incident on the eyelens.

OR

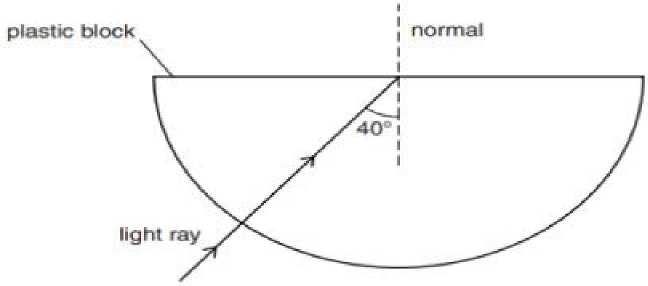
(B) Splitting of white light into its component colours.

Variation of refractive index of glass for different colours.



Dispersion of white light by a glass prism

36. (i) Explain why the refractive index of any material with respect to air is always greater 1.
(ii) In the figure below a light ray travels from air into the semi-circular plastic block. Give a reason why the ray does not deviate at the semi-circular boundary of the plastic block.

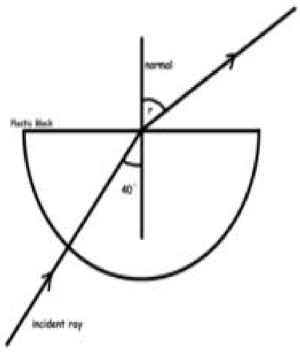


(iii) Complete the ray diagram of the above scenario when the light ray comes out of the plastic block from the top flat end. (3)

Ans: (i) The refractive index of a medium with respect to air is given by $\frac{speed\ of\ light\ in\ air}{speed\ of\ light\ in\ medium}$ speed of light in the medium is always less than the speed of light in air, hence the above ratio is always greater than 1.

(ii) The ray of light is undergoing normal incidence at the air-plastic block interface. And for normal incidence there is no deviation.

(iii)



37. Anannya responded to the question: Why do electrical appliances with metallic bodies are connected to the mains through a three pin plug, whereas an electric bulb can be connected with a two pin plug?

She wrote: Three pin connections reduce heating of connecting wires.

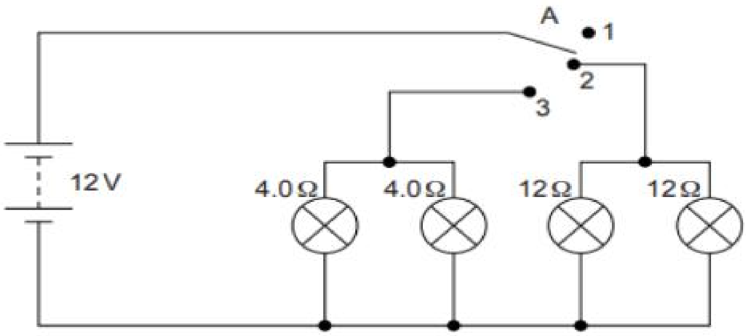
(i) Is her answer correct or incorrect? Justify.

(ii) What is the function of a fuse in a domestic circuit? (3)

Ans: (i) Anannya's answer is wrong. Electrical appliances with metallic bodies need an earth wire which provides a low resistance conducting path to the flow of current, in case there is an accidental leakage of current through the conducting body of the appliances.

(ii) An electrical fuse is a safety device that operates to provide protection against the overflow of current in an electrical circuit. An important component of an electrical fuse is a metal wire or strip that melts when excess current flows through it.

38.



Vinita and Ahmed demonstrated a circuit that operates the two headlights and the two sidelights of a car, in their school exhibition. Based on their demonstrated circuit, answer the following questions.

(i) State what happens when switch A is connected to

a) Position 2

b) Position 3

(ii) Find the potential difference across each lamp when lit.

(iii) Calculate the current

a) in each 12 Ω lamp when lit.

b) In each 4 Ω lamp when lit.

OR

(iv) Show, with calculations, which type of lamp, 4.0 Ω or 12 Ω, has the higher power.

(1+1+2)

Ans: (i) a) Only 12 Ω lamp is on.

b) Only 4Ω lamp is on.

(ii) 12 V for both sets of lamps when connected.

(iii) When 12 Ω lamps are on;

P.d. across each bulb is 12 V, hence current is 1A

When 4Ω lamps are on;

p.d. across each lamp is 12 V, hence current is 3A

OR

(iv) Power is given by $P = \frac{V^2}{R}$.

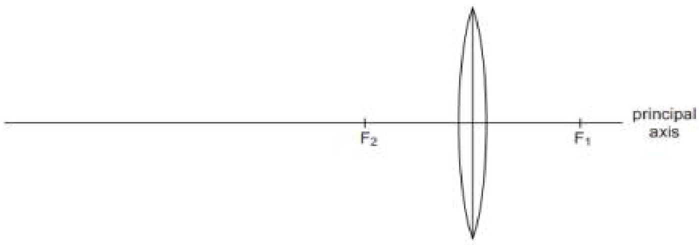
Voltage across each bulb is 12 V.

Power of 12Ω bulb $P = \frac{12 \times 12}{12} = 12\,W$

Power of 4Ω bulb $P = \frac{12 \times 12}{4} = 36\,W$

Power of 4 Ω bulb is greater.

39.



(A) The above image shows a thin lens of focal length 5m.
(i) What is the kind of lens shown in the above figure?
(ii) If a real inverted image is to be formed by this lens at a distance of 7m from the pole, then show with calculation where should the object be placed?
(iii) Draw a neatly labelled diagram of the image formation mentioned in (ii) (5)

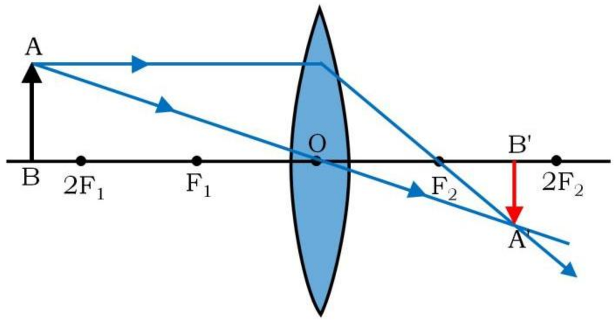
OR

(B) A 10 cm long pencil is placed 5 cm in front of a concave mirror having a radius of curvature of 40 cm.
(i) Determine the position of the image formed by this mirror.
(ii) What is the size of the image?
(iii)Draw a ray diagram to show the formation of the image as mentioned in the part (i). (5)

Ans: (A) (i) Convex lens

(ii) using lens formula $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}, \frac{1}{5} = \frac{1}{7} - \frac{1}{-u}$ **or** $u = -\frac{35}{2} m$

(iii)



OR

(B) (i) Using mirror formula; $\frac{1}{f} = \frac{1}{v} + \frac{1}{u}, \frac{1}{-20} = \frac{1}{v} + \frac{1}{-5}$ **or** $v = +\frac{20}{3} cm$

(ii) $m = \frac{v}{u} = \frac{20}{5\times 3} = \frac{4}{3}$**, size of image** $\frac{4}{3} \times 10 = \frac{40}{3} cm$

(iii)